

DATA ENGINEER

Name : Sreesai K

Contact : 952-295-5854

Email : sreesaik0912@gmail.com

URL: <https://www.linkedin.com/in/sreesai123/>

PROFESSIONAL SUMMARY :

- Over 10+ years of experience in Information Technology which includes Hands on experience in Hadoop ecosystem including Spark, Kafka, HBase, MapReduce, Python, Scala, Pig, Impala, Sqoop, Oozie, Flume, Storm, data technologies and worked on Spark SQL, Spark Streaming and using Core Spark API to explore Spark features to build data pipelines.
- Experience in building data solutions using Lambda Architecture using Cloudera distribution of Hadoop, Twitter Storm, Trident, MapReduce, Cascading, HIVE, PIG and Sqoop.
- Strong Knowledge and experience on implementing Data in Amazon Elastic MapReduce (Amazon EMR) for processing, managing Hadoop framework dynamically scalable Amazon EC2 instances, Lambda, SNS, SQS, AWS Glue, S3, RDS and Redshift.
- end-to-end implementation project experience on data processing pipeline using Hadoop (CDP) ecosystem, Data Lake, and Data Warehouse.
- Strong experience with AWS Cloud on data integration with Databricks, Apache Spark, Airflow, EMR, Glue, Kafka, Kinesis and Lambda in S3, Redshift, RDS, MongoDB/DynamoDB ecosystems.
- Responsible for the design and development of data related solutions (databases, ETL systems, code, scripts, data models, reports, documentation) and supporting various products. Extensive hands-on experience developing database queries, ETL, performance tuning and database unit tests in a relational database environment.
- Having experience in developing a data pipeline using Kafka to store data into HDFS and Experience in building Real-time Data Pipelines with Kafka Connect and Spark Streaming and developing end to end data processing pipelines that begin with receiving data using distributed messaging systems Kafka for persisting data into Hive.
- Experienced in development methodologies like Agile/Scrum.
- Excellent Experience in Designing, Developing, Documenting, Testing of ETL jobs and mappings in Server and Parallel jobs using Data Stage to populate tables in Data Warehouse and Data marts.
- Expertise in transforming business requirements into analytical models, designing algorithms, building models, developing Data Mining, Data Acquisition, Data Preparation, Data Manipulation, Feature Engineering, Machine Learning Algorithms, Validation and Visualization and reporting solutions that scales across massive volume of structured and unstructured Data.
- Strong knowledge in working with ETL methods for data extraction, transformation, and loading in corporate- wide ETL Solutions and Data Warehouse tools for reporting and data analysis
- Experience with different ETL tool environments like SSIS, Informatica, and reporting tool environments like SQL Server Reporting Services, and Business Objects.
- Having good experience on all flavors of Hadoop (Cloudera, Hortonworks, and MapR etc.) and Hands on experience in AVRO and Parquet file format, Dynamic Partitions, Bucketing for best Practice and Performance improvement.

- Expert in designing Server jobs using various types of stages like Sequential file, ODBC, Hashed file, Aggregator, Transformer, Sort, Link Partitioner and Link Collector.
- Proficiency in Data Practices and Technologies like HDFS, MapReduce, Hive, Pig, HBase, Sqoop, Oozie, Flume, Spark, Kafka.
- Develop data pipeline to synchronize and aggregate data from various data sources. Build the query data platform and provide guidelines for partner team to use aggregated data. Publish standard data sets & schema.
- Experience with source control systems such as Git, Bitbucket, and Jenkins. Working experience in CI/CD deployments
- Good knowledge in RDBMS concepts (Oracle 112c/1g, MS SQL Server 2012) and strong SQL, query writing skills (by using Erwin & SQL Developer tools), Stored Procedures and Triggers and Experience in writing Complex SQL Queries involving multiple tables inner and outer joins.
- Excellent knowledge with Unit Testing, Regression Testing, Integration Testing, User Acceptance Testing, Production implementation and Maintenance.
- Working knowledge of Azure cloud components (HDInsight, Databricks, Data Lake, Blob Storage, Data Factory, Storage Explorer, SQL DB, SQL DWH, Cosmos DB).
- Experienced in building data pipelines using Azure Data Factory, Azure Databricks, and loading data to Azure Data Lake, Azure SQL Database, Azure SQL Data Warehouse, and controlling database access.
- Extensive experience with Azure services like HDInsight, Stream Analytics, Active Directory, Blob Storage, Cosmos DB, and Storage Explorer.

Programming Languages Big Data Ecosystem	Java, Python, SQL, and C/C++ Hadoop, Map Reduce, Kafka, Spark, Pig, Hive, YARN, Flume, Sqoop, Oozie, Zookeeper, Talend.
Hadoop Distributions	Cloudera Enterprise, Horton Works, EMC Pivotal.
Databases	Oracle, SQL Server, PostgreSQL.
Web Technologies	HTML, XML, JQuery, Ajax, CSS, JavaScript, JSON.
Streaming Tools	Kafka
Testing	Hadoop Testing, Hive Testing, MRUnit.
Operating Systems	Linux Red Hat/Ubuntu/CentOS, Windows 10/8.1/7/XP.
Cloud	AWS EMR, Glue, RDS, Cloud Watch, S3, Redshift Cluster, Kinesis, Dynamo DB.
Technologies and Tools	Servlets, JSP, Spring (Boot, MVC, Batch, Security), Web Services, Hibernate, Maven, GitHub, Bamboo.

TECHNICAL SKILLS

Client: Huntington Bank, Remote
Present

April 2023 to

Role: Sr. Data Engineer

Responsibilities:

- Worked with all phases of Software Development Life Cycle and used Agile methodology for development.
- Involved in gathering requirements, design, implementation, deployment, testing and maintaining of the applications to meet the organization's needs

using SCRUM methodology.

- Participated in scrum meetings and coordinated with Business Analysts to understand the business needs and implement the same into a functional design.
- Used Azure Data Factory extensively for ingesting data from disparate source systems. Involved in Requirement gathering, business Analysis, Design and Development, testing and implementation of business rules.
- Understand business use cases, integration business, write business & technical requirements documents, logic diagrams, process flow charts, and other application related documents.
- Used Pandas in Python for Data Cleansing and validating the source data.
- Designed and developed ETL pipeline in Azure cloud which gets customer data from API and processes it to Azure SQL DB.
- Orchestrated all Data pipelines using Azure Data Factory and built a custom alerts platform for monitoring.
- Created custom alerts queries in Log Analytics and used Web hook actions to automate custom alerts.
- Created Databricks Job workflows which extracts data from SQL server and upload the files to sftp using pyspark and python. Design and implement database solutions in Azure SQL Data Warehouse, Azure SQL.
- Migrate data from traditional database systems to Azure databases.
- Design and implement migration strategies for traditional systems on Azure (Lift and shift/Azure Migrate, other third-party tools.
- Experience in DWH/BI project implementation using Azure Data Factory.
- Interacts with Business Analysts, Users, and SMEs on elaborating requirements.
- Design and implement end-to-end data solutions (storage, integration, processing, visualization) in Azure.
- Design and implement database solutions in Azure SQL Data Warehouse, Azure SQL.
- Propose architectures considering cost/spend in Azure and develop recommendations to right-size data infrastructure.
- Setup and maintain the Azure SQL Database, Azure Analysis Service, Azure SQL Data warehouse, Azure Data Factory, Azure SQL Data warehouse.
- Develop conceptual solutions & create proofs-of-concept to demonstrate viability of solutions.
- Implement Copy activity, Custom Azure Data Factory Pipeline Activities.
- Primarily involved in Data Migration using SQL, SQL Azure, Azure storage, and Azure Data Factory, SSIS, PowerShell.
- Create C# applications to load data from Azure storage blob to Azure SQL, to load from web API to Azure SQL and scheduled web jobs for daily loads.
- Recreating existing application logic and functionality in the Azure Data Lake, Data Factory, SQL Database and SQL Data Warehouse environment. experience in DWH/BI project implementation using Azure DF and data bricks. Architect, design and validate Azure infrastructure-as-a-Service (IaaS) environment.
- Develop dashboards and visualizations to help business users analyze data as well as providing data insight to upper management with a focus on Microsoft

- products like SQL Server Reporting Services (SSRS) and Power BI.
- Performed Application-level DBA activities creating tables, indexes, and monitored and tuned Teradata BETQ scripts using Teradata Visual Explain utility.
- Performance tuning, monitoring, UNIX shell scripting, and physical and logical database design.
- Developed UNIX scripts to automate different tasks involved as part of the loading process.
- Worked on Tableau software for the reporting needs.
- Worked on creating a few Tableau dashboard reports, Heat map charts and supported numerous dashboards, pie charts and heat map charts that were built on Teradata database.

Environment: Azure Cloud, Azure Data Factory (ADF v2), Azure functions Apps, Azure Data Lake, BLOB Storage, SQL server, Teradata Utilities, Windows remote desktop, UNIX Shell Scripting, AZURE PowerShell, Data bricks, Python, Erwin Data Modelling Tool, Azure Cosmos DB, Azure Stream Analytics, Azure Event Hub

Client: Johnson & Johnson, CA
March 2023

Jan 2020 to

Role: AWS Data Engineer

Responsibilities:

- Developed Spark applications using Pyspark and Spark-SQL for data extraction, transformation, and aggregation from multiple file formats.
- Design and Develop ETL Processes in AWS Glue to migrate Campaign data from external sources like S3, ORC/Parquet/Text Files into AWS Redshift
- Used Spark Streaming to receive real time data from the Kafka and store the stream data to HDFS using Python and NoSQL databases such as HBase and Cassandra
- Develop Spark streaming application to read raw packet data from Kafka topics, format it to JSON and push back to Kafka for future use case's purpose.
- Collected data using Spark Streaming from AWS S3 bucket in near-real-time and performs necessary Transformations and Aggregation on the fly to build the common learner data model and persists the data in HDFS.
- Designed and Developed ETL Processes in AWS Glue to migrate Campaign data from external sources like S3, ORC/Parquet/Text Files into AWS Redshift.
- Involved in file movements between HDFS and AWS S3 and extensively worked with S3 bucket in AWS.
- Experience with setting up and managing Databricks clusters for data processing and analysis.
- Proficient in working with Databricks notebooks to develop, test, and deploy ETL pipelines using languages such as Python, Scala, or SQL.
- Strong proficiency in Scala, with experience using it to build data processing pipelines, ETL workflows, and other data engineering tasks.
- Experience with AWS services related to data processing, such as Amazon EMR, Amazon S3, and AWS Glue, as well as proficiency in scripting languages such as Python and Bash for data processing and automation tasks.
- Create Snapshots of EBS Volumes and Monitor AWS EC2 Instances using Cloud Watch and worked on AWS Security Groups and their rules.

- Developed calculated columns and measures using complex DAX functions and power pivot.
- Validated the test data in DB2 tables on Mainframes and on Teradata using SQL queries.
- Knowledge of configuring Databricks jobs and scheduling them to run on a regular basis to ensure the timely processing of data.
- Expertise in integrating Databricks with other tools such as Apache Spark, Apache Hadoop, and cloud services like AWS.
- Authoring Python (PySpark) Scripts for custom UDF's for Row/ Column manipulations, merges, aggregations, stacking, data labeling and for all Cleaning and conforming tasks.
- Worked on AWS CLI Auto Scaling and Cloud watch monitoring creation and update.
- Handled AWS Management Tools as Cloud watch and Cloud Trail.
- Stored the log files in AWS S3. Used versioning in S3 buckets where the highly sensitive information is stored.
- Integrated AWS Dynamo DB using AWS lambda to store the values of items and backup the DynamoDB streams.
- Prepared scripts to automate the Ingestion process using Pyspark and Scala as needed through various sources such as API, AWS S3, Teradata and Redshift.
- Used SQL Server Management Tool to check the data in the database as compared to the requirement given.
- Played a lead role in gathering requirements, analysis of entire system and providing estimation on development, testing efforts.
- Involved in designing different components of system like Sqoop, Hadoop process involves map reduce & hive, Spark, FTP integration to down systems.
- Migrated an existing on-premises application to AWS. Used AWS services like EC2 and S3 for small data sets processing and storage, Experienced in Maintaining the Hadoop cluster on AWS EMR

Environment: Python, Spark, AWS EC2, AWS S3, AWS EMR, AWS Redshift, AWS Glue, AWS RDS, AWS SNS, AWS SQS, AWS Athena, Snowflake, Data warehouse, Airflow, Data Governance, Kafka, ETL, Terraform, Docker, SQL, Tableau, Git, REST, Bitbucket, Jira.

Client: State of IDAHO

Nov

2018 to Dec 2019

Role: Sr. Data Engineer

Responsibilities:

- Extensive experience in designing, implementing, and managing data solutions on AWS, leveraging services such as Amazon S3, Amazon Redshift, Amazon Glue, and Amazon EMR.
- Proficient in building and optimizing scalable data pipelines using AWS Data Pipeline, AWS Glue, and Databricks to extract, transform, and load data from various sources.
- Hands-on expertise in implementing big data processing and analytics workflows using Databricks, leveraging Apache Spark for data ingestion, transformation, and analysis.
- Strong understanding and proficient in writing Spark jobs using Scala or

Python, leveraging Databricks notebooks for data exploration, feature engineering, and model development.

- Experienced in building and managing Spark clusters on AWS Databricks, including cluster provisioning, configuration, and performance optimization.
- Implemented real-time data processing and streaming analytics solutions using AWS services such as Amazon Kinesis and Databricks Structured Streaming.
- Developed Spark applications using Scala for easy Hadoop transitions.
- Used Spark and Spark-SQL to read the parquet data and create the tables in hive using the Scala API.
- Performed advanced procedures like text analytics and processing, using the in-memory. computing capabilities of Spark using Scala.
- Developed Spark code using Scala and Spark-SQL for faster processing and testing.
- Used Spark-Streaming APIs to perform necessary transformations and actions on the fly.
- for building the common learner data model which gets the data from Kafka in near real time.
- Responsible for loading Data pipelines from web servers and Teradata using Sqoop with Kafka and Spark Streaming API.
- Implemented data warehousing solutions on AWS using Amazon Redshift, including schema design, SQL script development, and query optimization.
- Experience in deploying and managing machine learning models on AWS using Databricks and Amazon SageMaker, integrating them with data pipelines for real-time predictions.
- Implemented data governance practices on AWS, including data security, privacy, and compliance with regulations such as GDPR and HIPAA.
- Proficient in leveraging AWS Lambda for serverless data processing, event-driven architectures, and data transformation tasks.
- Experience in implementing data lakes on AWS using Amazon S3 for scalable storage and processing of structured and unstructured data.
- Implemented data replication and synchronization solutions using AWS services such as AWS Data Sync and AWS Glue to ensure data consistency across multiple environments.
- Implemented data security measures on AWS, including encryption at rest and in transit, identity and access management (IAM), and network security controls.
- Proficient in deploying and managing ETL processes on AWS using AWS Glue, leveraging its data discovery, schema evolution, and data transformation capabilities.

Environment: Agile Scrum, Spark, Scala, Hive, Kafka, Python, AWS (EC2, S3, EBS, ELB, RDS, SNS, SQS, VPC, Cloud formation, CloudWatch, ELK Stack), Bitbucket, Ansible.

Client: BIG BASKET, Hyderabad
August 2018

June 2014 to

Role: Big Data Developer

Responsibilities:

- Implementing the Proof of Concept (POC) for ETL Abinitio graph concepts which need to be migrated into Spark using scala and python (Pyspark).
- Develop Data pipelines using Sqoop, Spark, and Map reduce and Hive to Ingest, transform and analyze customer behavior data.
- Developed a data pipeline using Kafka, Spark Streaming and Hive to ingest the data from data lakes to Hadoop distributed file system.
- Implemented Spark using python and Spark SQL for faster processing of data and algorithms for real time analysis in Spark.
- Used Spark for interactive queries, processing of streaming data and integration with popular NOSQL database for huge volume of data.
- Involved in converting Hive/SQL queries into Spark transformations using Spark RDDs and Python.
- Handled importing data from different data sources into HDFS using Sqoop and also performing transformations using Hive, Map Reduce and then loading data into HDFS.
- Extracting the Files from the RDBMS (DB2) by using Sqoop into Hadoop file system (HDFS) to process the workflow.
- Implementing the Partitioning and bucketing for faster query processing in Hive Query Language (HQL).
- Involving in Converting the HIVE/SQL queries into Spark transformation using Spark Data frames, Datasets and User defined functions (UDF's).
- Design Hive queries and Pig Script to perform Data Analysis, Data transfer and Table design.
- Evaluating the Data between the ETL and Hadoop to ensure Data quality.
- Responsible in creating mappings and workflows to extract and load data from relational databases, flat file sources and legacy systems.
- Testing on Apache Tez Framework and Hadoop Map Reduce frameworks for building high performance batch and interactive Data Processing Applications.
- Reconciling the data on daily basis in between the ETL and Hive tables by using a compare tool which is implemented in spark framework using Pyspark.
- Fine tune Hadoop applications for high performance and throughput, troubleshoot and debug any Hadoop ecosystem run time issues.
- Performing Data validation operation between ETL and Apache Hive tables.
- Developing the Linux shell scripting for Deploying and running the migrated Hadoop Application in Production Servers.
- Developing Workflows for scheduling and orchestrating the Hadoop Process.

Environment: Hadoop, Spark, Scala, MapReduce, HDFS, Hive, Java, SQL, Cloudera Manager, Sqoop, Oozie, Zookeeper, Apache Pig, Python, Java, SSRS, HBase, Cassandra, PySpark, Apache Kafka, HIVE, FLUME, ETL, UDF.